TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
System of inequalities: graph and shade each one; the overlap is the solution set.
A solution makes EVERY inequality true. Points on a dashed line are not solutions.
In a story: no negative amounts, so stay in the first quadrant.

1. Is (2, 1) a solution of $y \geq x - 3$ and $y < -x + 5$?

Tip: Test the point in EVERY inequality.

A. No, only the first is true

B. No, only the second is true

C. Yes, it makes both true

2. A raft holds at most 600 lb. Adults weigh about 150 lb and children 75 lb. Adults pay $40 and children $20, and the guide wants at least $100. Which system models this?

Tip: One inequality for weight, one for money.

F. $150a + 75c \geq 600$ and $40a + 20c \leq 100$

G. $40a + 20c \leq 600$ and $150a + 75c \geq 100$

H. $150a + 75c \leq 600$ and $40a + 20c \geq 100$

3. Name one point that is a solution of $y > x$ and $y < 4$, and show that it works.

> **Tip:** Pick a point, then test BOTH inequalities.

My point: (____, ____)

y > x: _____ ✓

y < 4: _____ ✓